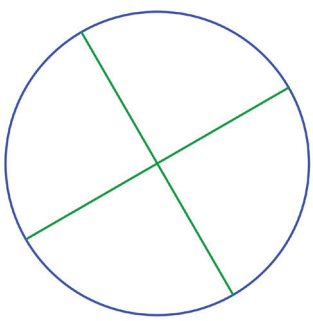
10 Circles And Lines

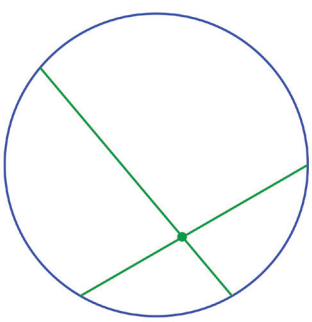
Chords

Any two diameters of a circle intersect at the centre of the circle:

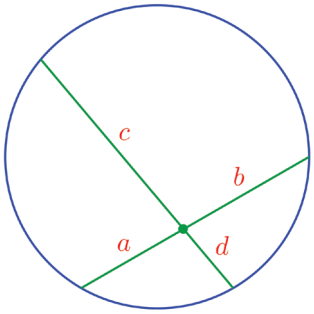


All four parts made by the intersection are equal to the radius of the circle.

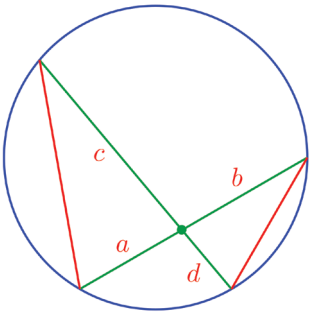
What happens when two chords, which are not diameters, intersect?



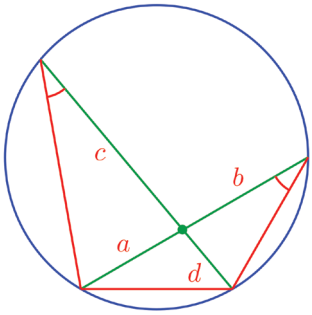
The parts may not be equal. But there is a relation between them. To see it, let's denote the lengths of the parts of one chord as *a, b* and the lengths of the parts of the other as *c, d*:

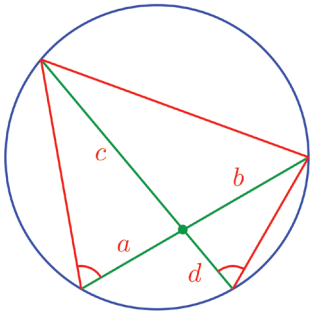


To find the relation between the parts, we join the ends of the chords to make two triangles as shown below:

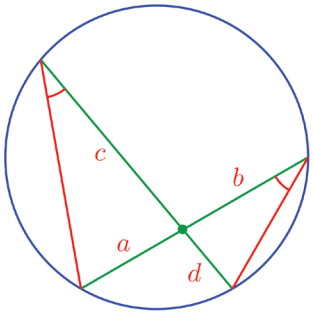


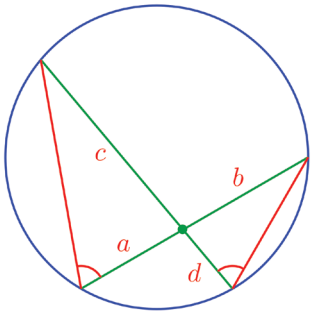
The top angles of the triangles are angles in a segment of the circle; and the bottom angles are angles in another segment:





So the top angles of the triangles in the first picture are equal; and so are the bottom angles:





This means the triangles have the same angles. So, the lengths of the sides opposite the equal angles are scaled by the same factor (the section **Angles and sides** of the lesson **Similar Triangles** in the Class 9 textbook).

The lengths of the sides opposite the top angles are a and d; and the lengths of the sides opposite the bottom angles are c and b.

They are scaled by the same factor. If we denote the scale factor as k, then we get

$$d = ka$$

$$b = kc$$

We don't know the value of k. To eliminate k and get a relation between $a, b, c, d,$ we first calculate from the first equation

$$k = \frac{d}{a}$$

Using this in the second equation, we get

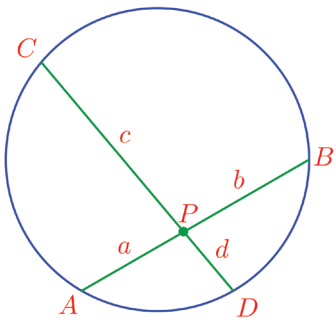
$$b = \frac{d}{a} \times c = \frac{cd}{a}$$

We can rewrite this as

$$ab = cd$$

If we name the chords AB and CD and their point of intersection P, we can write this as

$$AP \times PB = CP \times PD$$



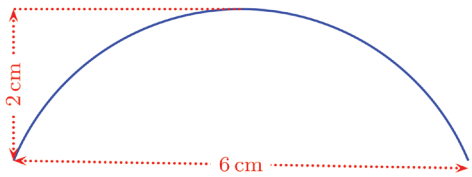
Draw a circle and two chords AB and CD in GeoGebra. Mark the point P where the chords intersect. If we type PA * PB as Input, we get the value of PA × PB in the Algebra view. Compute PC × PD also like this. Compare these products. What happens to these when P is on the circle?

We can state this as a general result:

When two chords of a circle intersect within the circle, the product of the parts of one chord is equal to the product of the parts of the other

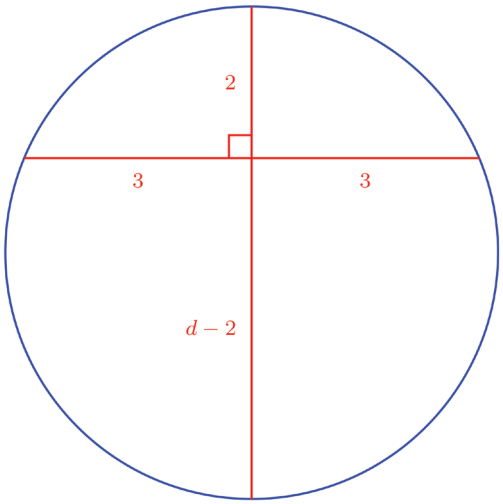
Let's look at a problem using this:

The picture shows a part of a ring:



Calculate the diameter of the ring.

Let's imagine the full circle, the chord joining the ends of this arc and the diameter perpendicular to it. This diameter bisects the chord, since it is perpendicular to the chord. So taking the diameter as d centimetres, we get the picture below:



Now using the principle stated above, we get

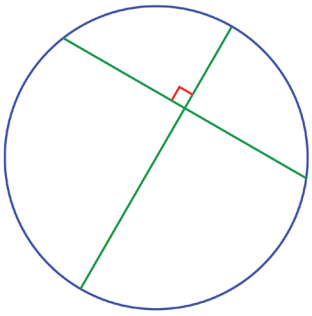
$$2(d-2) = 3 \times 3 = 9$$

From this we get

$$d = \frac{9}{2} + 2 = 4\frac{1}{2} + 2 = 6\frac{1}{2}$$

Thus the diameter of the ring is $6\frac{1}{2}$ centimetres.

In this problem, one of the intersecting chords is a diameter of the circle and the other is a chord perpendicular to it. See such a picture:



Lotus problem

We have mentioned the book *lilavati* of Bhaskaraacharya in the Class 8 textbook (the section *Inverse operations* of the lesson *Solutions of Equations*. The translation of a *sloka* in it is this:

> In a lake full of frolicking chakravaka and krauncha birds, a lotus bud stands erect, half a hand-span high. Swaying gently in the breeze, it submerged, two hand-span away. Oh reckoner!, quickly tell the depth of the lake.

Try to find the answer of this question, as you found the diametre of the ring:

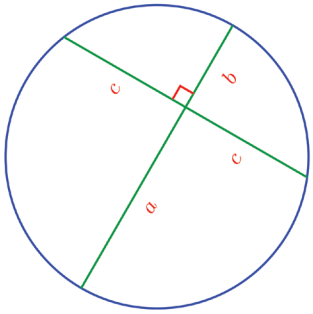
The actual sloka is:

chakrakraunchaakulitasalilee
kvaapidrshtam tadaagee
toyaaduurdhva kamalakalikagram
vitasti pramaanam
mandam mandam chalitamanileenaahatam
hastayugme
tasminmagnam ganaka, kathaya
kshipramambupramaanam

Let's denote the lengths of the parts of the diameter as a and b.

Since the diameter is the perpendicular from the centre of the circle to a chord, it bisects the chord (the section **Chords** of the lesson **Circles** in the Class 8 textbook).

So, we can denote the lengths of both parts of the chord as c:



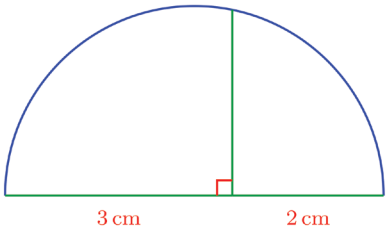
What is the relation between a, b and c according to our general principle?

$$ab = c^2$$

This we can state like this:

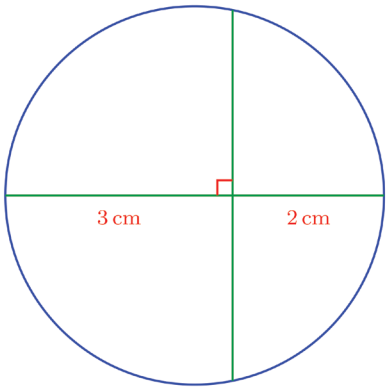
> The product of the parts into which a diameter of a circle is cut by a perpendicular chord, is the square of half the chord

For example, see this picture:



The perpendicular from a point on the diameter meets the semicircle. What is the length of this perpendicular?

If we draw the complete circle and extend the perpendicular downwards, it becomes a chord and the perpendicular is half the chord:



So, by the principle stated above, the square of the length of the perpendicular is $3 \times 2 = 6$; and the length of the perpendicular is $\sqrt{6}$ centimetres.

Like this, can't we draw lines of length as the square root of any number?

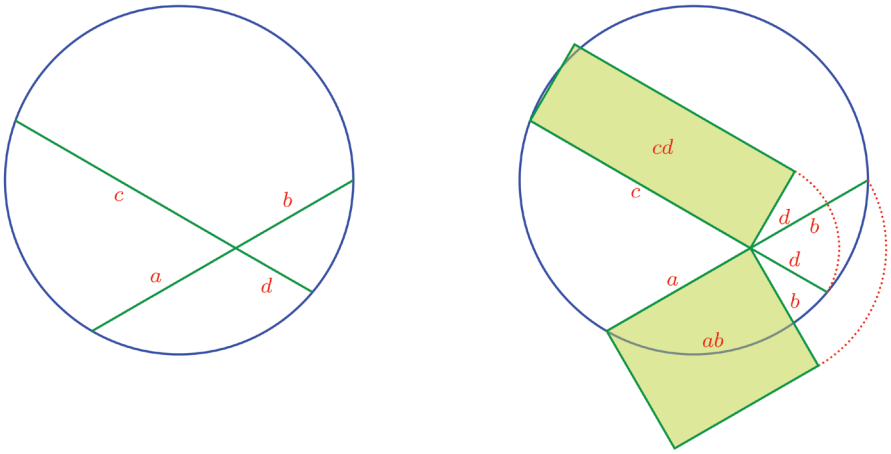
How do we draw a line of length $\sqrt{5}$ centimetres?

Can you draw a line of length $\sqrt{6}$ centimetres in any other way?

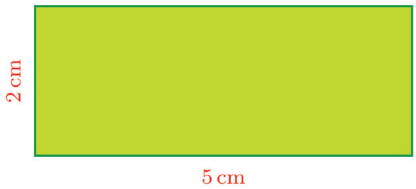
Areas

We have been talking about the products of the parts of two intersecting chords in a circle. The product of lengths can be considered as area. So the general principle about this can be stated like this:

When two chords intersect within a circle, the rectangles with sides as the parts of each chord have equal areas



We can use this to change one side of a rectangle without affecting the area. For example, see this rectangle:

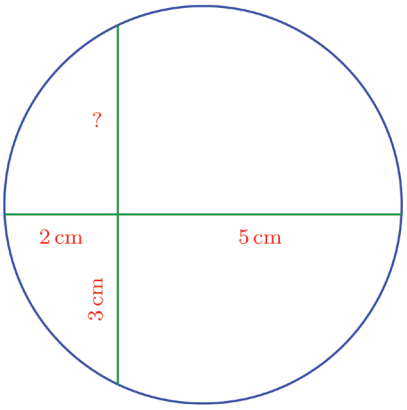


We want to draw a rectangle of the same area, with one side 3 centimetres.

The area of the rectangle is $5 \times 2 = 10$ square centimetres. If one side is to be 3 centimetres, the other should be $\frac{10}{3} = 3\frac{1}{3}$ centimetres. We cannot measure this exactly with a ruler, can we?

How can we use the result stated above?

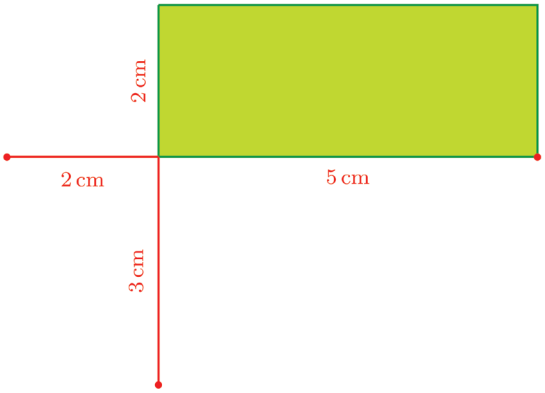
Imagine a pair of chords intersecting within a circle like this: the parts of one chord are 5 and 2 centimetres long, and one part of the other chord is 3 centimetres long. What would be the length of the other part?



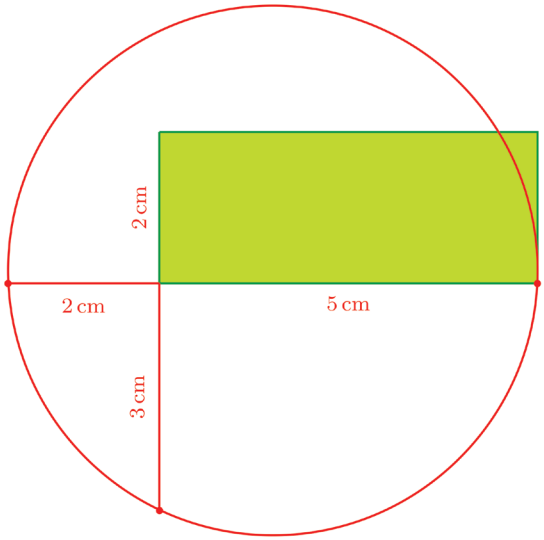
The length of this remaining part multiplied by 3 must be $2 \times 5 = 10$, right?

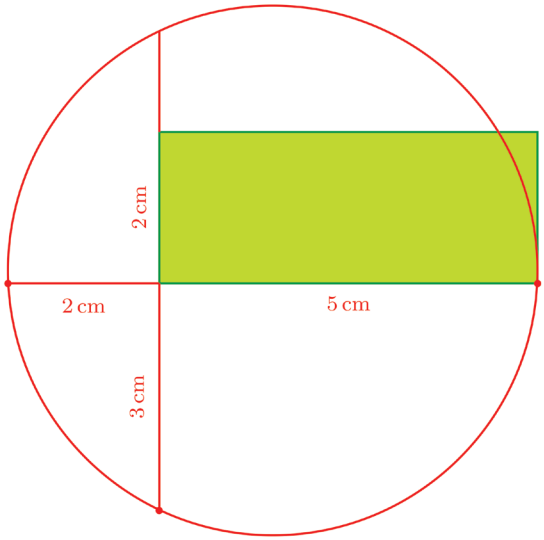
So this is the $3\frac{1}{3}$ centimetres long line we need. How do we actually draw it?

First let's extend the bottom side of the rectangle by 2 centimetres to the left and the left side downward by 3 centimetres:

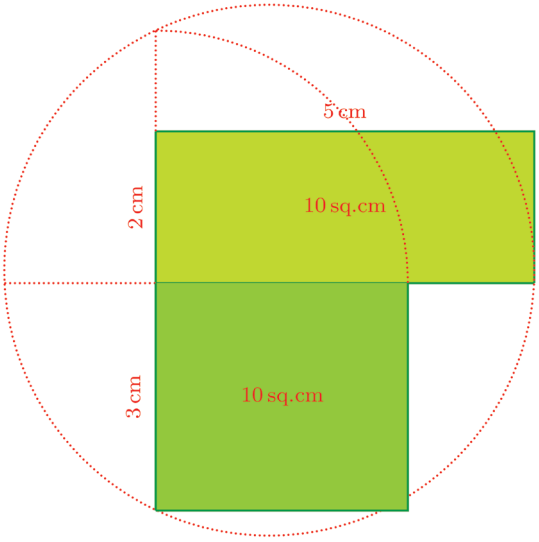


Next we draw the circle through the three points, left, right and down (we have seen how a circle can be drawn through three points in the section **Circumcircle** of the lesson **Circles** in the Class 8 textbook); and extend the left side of the rectangle to meet this circle.





Now the length we thus get can be marked horizontally to draw the rectangle we want:

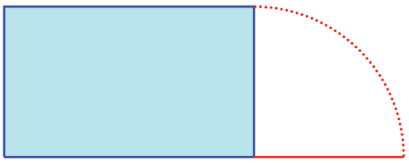


To draw such rectangles, we need not know the lengths of the sides of the original rectangle; only how much a side has to be lengthened or shortened.

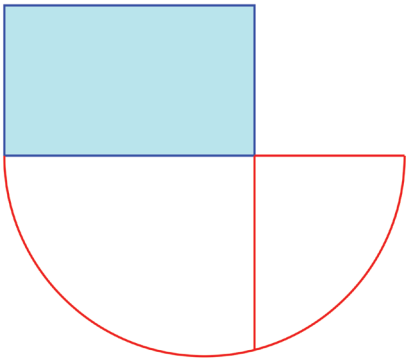
The corollary about the parts of a diameter cut by a perpendicular chord can be used to draw a square of the same area as a rectangle.

For example, see this rectangle:

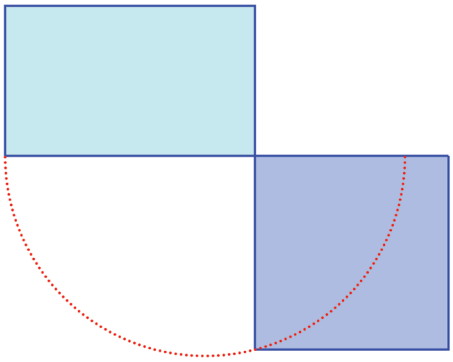
To draw a square of the same area as this, first extend the width of the rectangle by its height:



Now draw a semicircle downward with the bottom line as the diameter, and extend the right side of the rectangle downward to meet it:

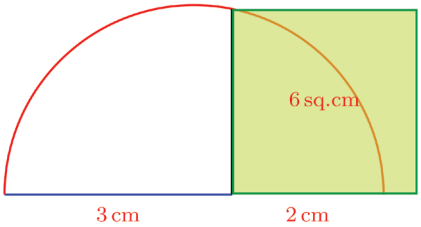


This line is the side of the square we want (why?)



Note that here also, we don't need the lengths of the sides of the rectangle.

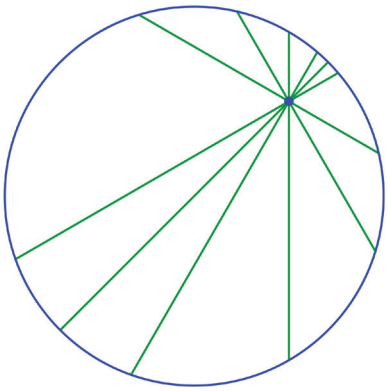
We can also use this idea to draw a square of specified area. For example, we saw in the last section how we can draw a line of length $\sqrt{6}$ centimetres using this. With this as the length of a side, we can draw a square of area 6 square centimetres:



(1) Draw a rectangle of width 5 centimetres and height 4 centimetres.
   (i) Draw a rectangle of the same area with width 6 centimetres.
   (ii) Draw a square of the same area.
(2) Draw a square of area 15 square centimetres.
(3) Draw a square of area 5 square centimetres in three different ways (Hint: Recall Pythagoras Theorem).

Line and point

See this picture:

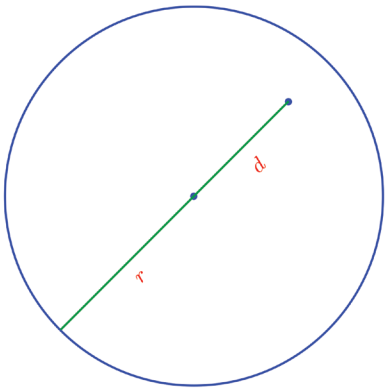


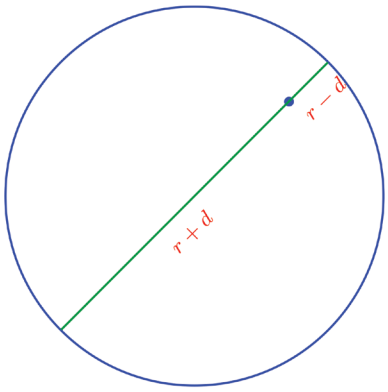
Draw a circle in GeoGebra and mark points A on the circle and P inside the circle. Select the Line tool and draw AP. Mark the point B where this line intersects the circle again. Hide the line and draw the chord AB. Compute PA x PB. Move A along the circle. Does the value of PA x PB change? What if we change the position of P? What happens to the product as we move P away from the centre of the circle? What if P is on the circle? When it is outside the circle?

Some chords pass through a single point within the circle. The point divides each chord into two parts. According to the general principle we have been using now, the product of the lengths of each pair of these parts gives the same number.

To calculate this number, let's draw the diameter through this point. Since it is also a chord of the circle, the product of the lengths of the parts into which the point divides the diameter is also equal to this number.

If we denote the radius of the circle as r and the distance of the point from the centre of the circle as d, we can write the lengths of the parts of the diameter like this:

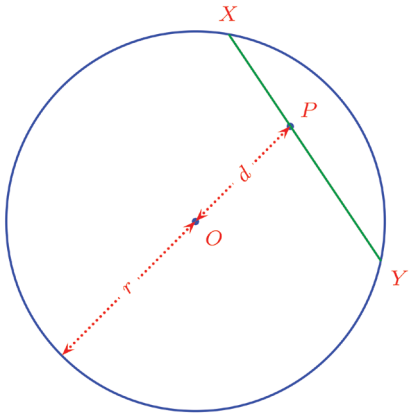




So, the product of these parts is

$$(r + d)\,(r - d) = r^2 - d^2$$

We have seen that the products of the lengths of the parts of any chord through the point is the same number. Thus the product of the lengths of the parts of every such chord is $r^2 - d^2$.



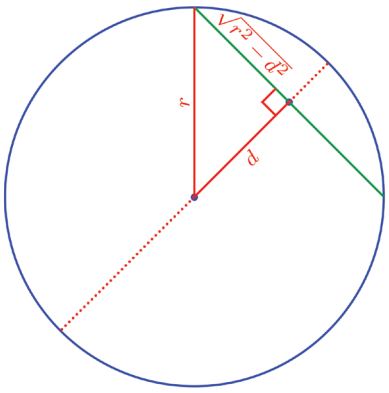
$$PX \times PY = r^2 - d^2$$

Thus we have this general result:

For any chord XY of a circle of radius r, passing through a point P within the circle at a distance d from the centre,

$$PX \times PY = r^2 - d^2$$

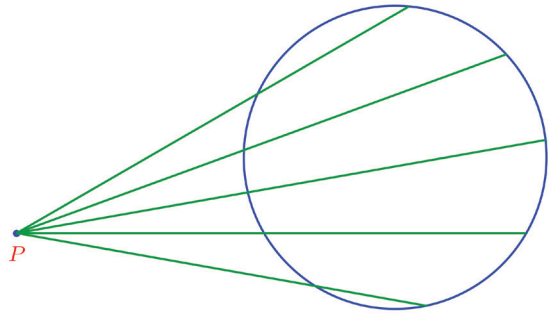
Geometrically, $r^2 - d^2$ is equal to the square of half the chord through P, perpendicular to the diameter through P.



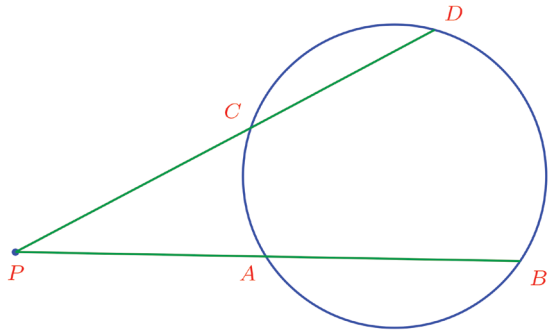
In the result above, P is a point within the circle and X, Y are points where a line through P intersects the circle, right? And PX, PY are the distances of these points from P.

What if we take a point outside the circle and draw lines intersecting the circle?

For such lines also, is the product of the distances from P to the points of intersection with the circle the same number?

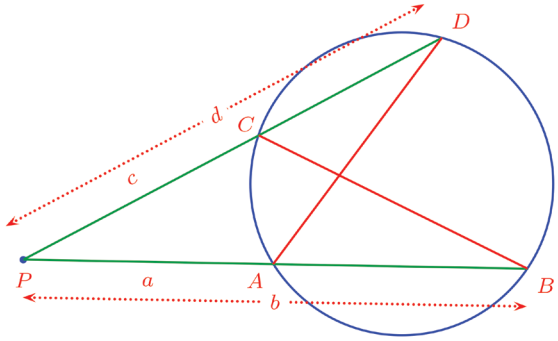


Let's take two such lines and check this:



We want to check whether the products $PA \times PB$ and $PC \times PD$ give the same number.

For that we draw lines to make triangles as in the picture below. We denote the lengths PA and PB as a and b, and the lengths PC and PD as c and d:



The triangle PAD and PCB have the same angle at P. The angles at B and D are equal (why?)

Thus these triangles have the same angles and so sides opposite to equal angles are scaled by the same factor.

In these triangles the lengths of the sides opposite to the angles at B and D are c and a; and the lengths of the sides opposite to the angles at C and A are b and d.

They are scaled by the same factor. Denote the scale factor as k, then we get

$$c = ka$$

$$b = kd$$

From this we get, as in the case of a point within the circle,

$$ab = cd$$

(Do it!)

That is,

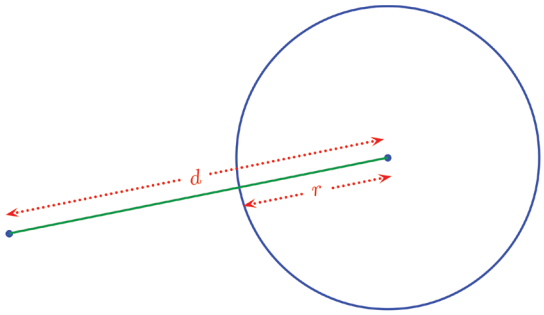
$$PA \times PB = PC \times PD$$

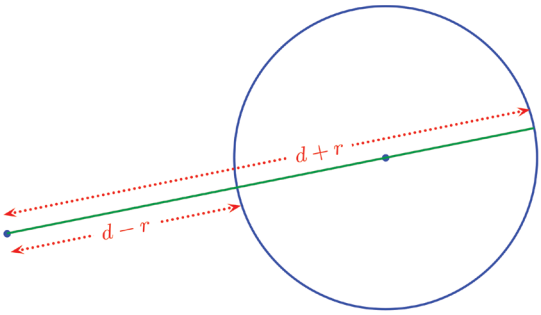
So, what do we see here?

For all lines through a point outside a circle intersecting the circle at two points, the product of the distances from the point to the points of intersection with the circle is the same number.

Can we calculate this number also using the radius of the circle and the distance of the point from the centre of the circle?

Let's denote the radius of the circle as r and the distance of the point from the centre of the circle as d. We take the line joining the point to the centre of the circle to compute the product of the distances:

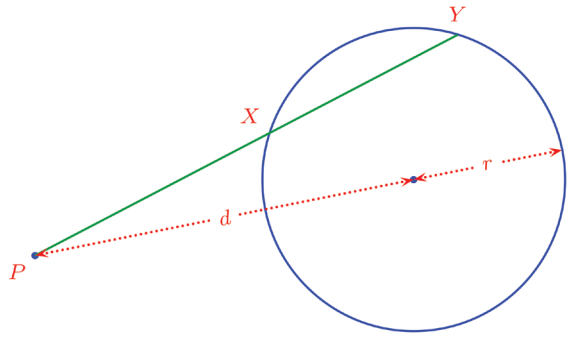




Here the distances from the point to the points of intersection with the circle can be seen as $d - r$ and $d + r$; and so their product is

$$(d + r)(d - r) = d^2 - r^2$$

Whatever line we draw through this point, the product of the distances from the point to the points of intersection with the circle would be this, isn't it?



$$PX \times PY = d^2 - r^2$$

Draw a circle in GeoGebra and mark a point P outside the circle. Draw the tangent from P to the circle and mark the point of contact T. Mark a point Y on the circle and draw the line PY. Mark the point X where this line meets the circle again. Compute PT^2 (for this type PT^2 as Input) and PX × PY. Compare these two numbers. Change the position of P and check.

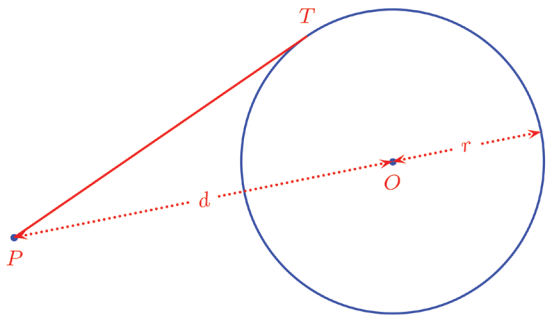
If a line from a point P outside a circle of radius r at a distance d from the centre cuts the circle at X and Y, then

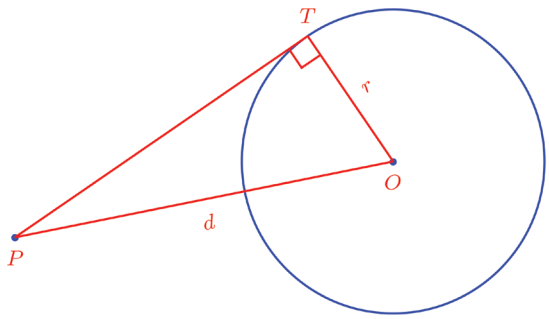
$$PX \times PY = d^2 - r^2$$

If P is a point inside the circle, we have seen that the product $PX \times PY$ can be geometrically seen as the square of half the length of a certain chord through P.

Is there such a geometric meaning if P is outside the circle?

To see this, draw the tangent from P to the circle and mark the point of contact as T. Joining T and P with the centre of the circle, we get a triangle:



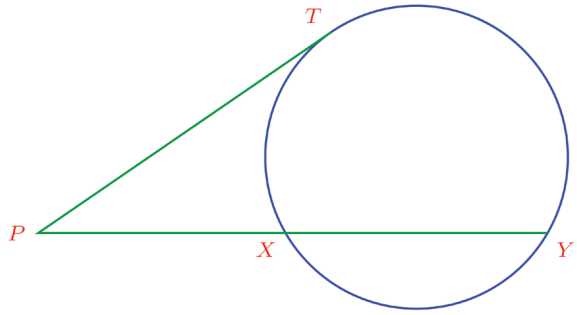


Since this is a right triangle (why?)

$$d^2 - r^2 = PT^2$$

Thus we have this result

Whatever line we draw from a point outside a circle, intersecting the circle at two points, the product of the distances from the point to the points of intersection is the square of the length of the tangent from the point.



$$PX \times PY = PT^2$$

Now let's compare the results about the lines drawn through a point inside or outside a circle

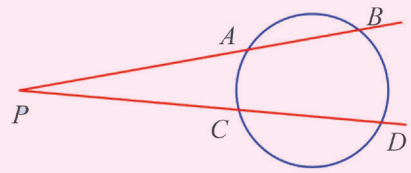
If we want to select a point at a distance d inside a circle of radius r, the distance d must be shorter than the radius r; and in this case the product of the distances from P to the points of intersection of a line through P with the circle is $r^2 - d^2$. If the point is to be outside the circle, d must be longer than the radius r; and the product as above is $d^2 - r^2$.

Whether the point is inside or outside the circle, what we do here is to subtract the larger of r^2 and d^2 from the smaller; thus in both cases we take the absolute value of $r^2 - d^2$ (the section **Distances** of the lesson **Real Numbers** in the Class 9 textbook).

So, we can say this in general:

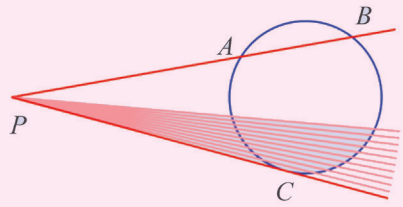
Unchanging relations

See this picture:



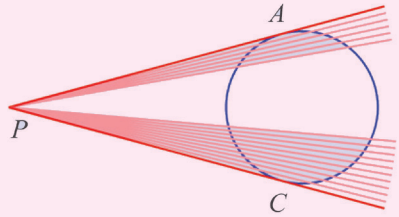
We know that $PA \times PB = PC \times PD$.

What if the bottom lines turns to become the tangent at C?



PD and PC coincide; and the relation between lengths becomes $PA \times PB = PC^2$

Suppose now the top line also turns to become the tangent at A:



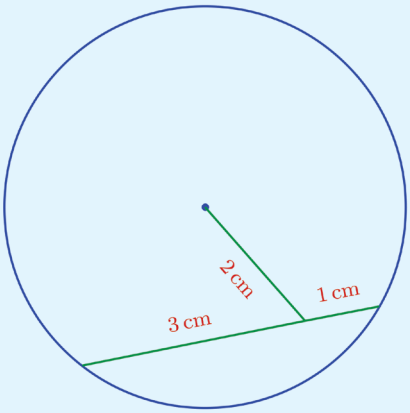
The relation becomes, $PA^2 = PC^2$; that is $PA = PC$.

We have already seen that the lengths of the tangents from a point are equal, haven't we?

If X and Y are the points of intersection of a circle of radius r with a line through a point P inside or outside the circle, at a distance d from the centre, then

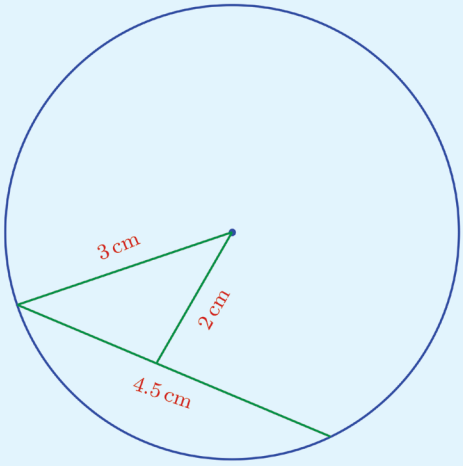
$$PX \times PY = |r^2 - d^2|$$

(1) In the picture, a line from the centre of a circle cuts a chord into two parts:



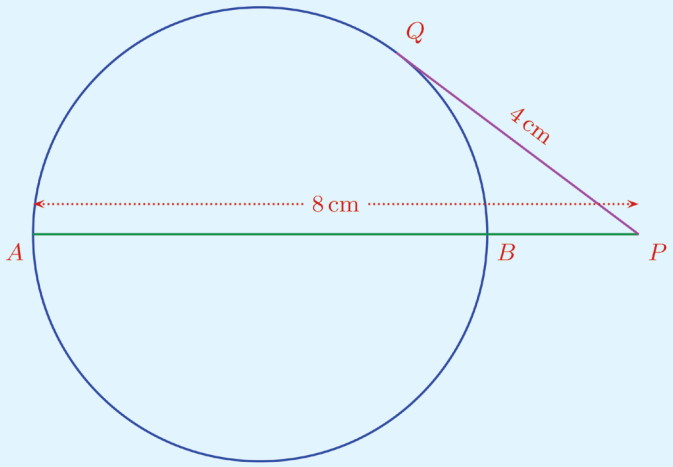
What is the radius of the circle?

(2) In the picture, a line from the centre of a circle meets a chord:



Find the lengths of the two parts of the chord.

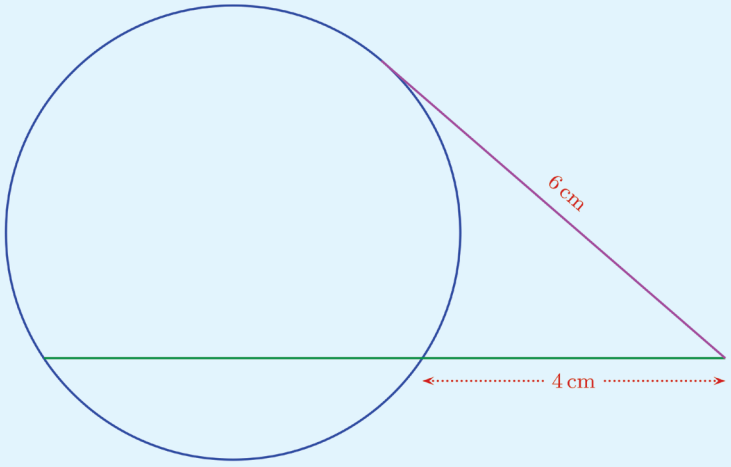
(3) In the picture, AB is a diameter of the circle and it is extended to a point P. The tangent from P touches the circle at Q:

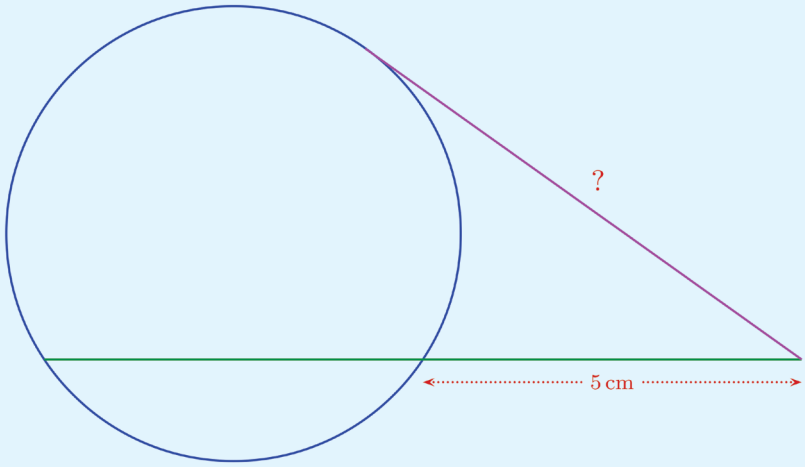


What is the radius of the circle?

(4) In the first of the two pictures below, the line joining two points of a circle is extended outward to a point and then a tangent is drawn from this point to the circle.

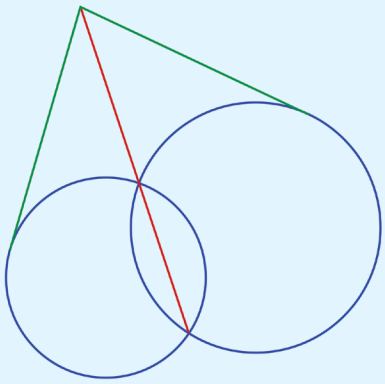
In the second picture, the same line is extended a bit more to a point and a tangent is drawn from this point:





What is the length of this tangent?

(5) In the picture, the line joining the points of intersection of two circles is drawn and from a point on it, one tangent is drawn to each circle:



Prove that the lengths of these tangents are equal.